

# Steven Golovkine

PHD IN APPLIED MATHEMATICS, STATISTICS

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## Current Position

### Assistant Professor (Professeur Adjoint)

UNIVERSITÉ LAVAL

- Develop novel methodologies for the modelling of functional data.
- Conduct lectures and tutorials for undergraduate and postgraduate students.
- Write research funding proposals.

Québec, Canada

Mar. 2025 - Present

## Experience

### Independent Researcher

NO AFFILIATION

- Conduct research on functional data analysis.
- Software: Python.

Toulouse, France

Jan. 2025 - Feb. 2025

### Postdoctoral Researcher on the FAST project

UNIVERSITY OF LIMERICK

- Develop novel, computationally efficient statistical models and algorithms for the modelling of multivariate sensor data.
- Demonstrate increased computational efficiency, so that the methodologies are applicable to modern, large-scale datasets.
- Supervise postgraduate students.
- Supervisor: Norma Bargary (UL) and Andrew Simpkin (UG)
- Software: Python

Limerick, Ireland

Feb. 2022 - Dec. 2024

### Data Scientist on the EMPOWER project

IRMES (INSEP)

- Maximize the performance of elite female athletes by optimizing their training responses with adapted workloads in synergy with their physiology and menstrual cycle.
- Develop statistical models to determine athletes hormonal profiles and analyze responses to training and competition loads.
- Link: <https://labos-recherche.insep.fr/fr/empowher>
- Software: **R**.

Paris, France

Jun. 2021 - Jan. 2022

### Research Engineer

TECHNOCENTRE, RENAULT

- Develop clustering methods for the analysis of autonomous vehicle Advanced Driver-Assistance Systems data.
- Create a Python package for Functional Data Analysis: FDApy.
- Use of Google Cloud Platform (Compute Engine and BigQuery) for the analysis of vehicle data.
- Software: Python and **R**.

Guyancourt, France

Jan. 2018 - Mar. 2021

### Data Scientist (Intern)

TECHNOCENTRE, RENAULT

- Implement a massive data analysis methodology for the validation of driving assistance systems.
- Software: Matlab.

Guyancourt, France

Apr. 2017 - Oct. 2017

### Research Assistant

COLORADO STATE UNIVERSITY

- Compare survey estimators for the *National Survey of College Graduation*.
- Software: **R**.

Fort Collins, USA

Jun. 2016 - Aug. 2016

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## Education

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### PhD. in Applied Mathematics, Statistics

TECHNOCENTRE, RENAULT AND CREST (ENSAI)

Guyancourt, France

Jan. 2018 - Jun. 2021

- **Title:** Statistical methods for multivariate functional data
- **Supervisors:** Valentin Patilea (Ensaï, CREST), Nicolas Klutchnikoff (Univ Rennes, IRMAR)
- **Funding:** Partnership with Groupe Renault through a CIFRE convention (100000€).
- **Abstract:** The topic of this thesis is related to functional data analysis and is motivated by modern data from automobile industry. The standard functional data methods rely on the assumption that the curves are continuously observed, without error. However, in general, the real data is neither continuously nor exactly observed. Therefore, a crucial step is to recover the trajectories from noisy measurements at discrete random points. For that, we propose an original point of view: the local regularity of the process generating the curves. Thus, combining information both within and across trajectories, we propose a simple estimator for this local regularity. Given this estimate, we build a nearly optimal local polynomial smoother of the curves from a sample of noisy trajectories. Nonparametric estimators for the mean and the covariance functions of functional data, using the local regularity of the process, are derived. Moreover, we propose a model-based clustering algorithm for a general class of functional data for which the components could be curves or images. Results of both simulated and real data show the good performances of this method. A Python package, implementing the methods and publicly available, has been developed.
- **Keywords:** Adaptive optimal smoothing, functional data analysis, gaussian mixtures, Hölder exponent, local polynomials, model-based clustering, multivariate functional principal component analysis, traffic flow.
- **Defense:** June 18, 2021 in front of the jury composed of André Mas (Professor, Montpellier University, President), Sophie Dabo-Niang (Professor, Lille University, Reviewer), Alois Kneip (Professor, Bonn University, Reviewer), Vincent Feuillard (Statistical Expert, Renault, Examiner) and Claire Gormley (Professor, University College Dublin, Examiner).
- **Manuscript:** tel-03540827

### MSc in Big Data

ENSAI (NATIONAL SCHOOL FOR STATISTICS AND DATA ANALYSIS)

Rennes, France

Sept. 2016 - Oct. 2017

- Dual degree program with Ensai engineering degree.
- Main topics: Statistics, Applied Mathematics, Computer Science.
- Training topics: assess, treat, and analyze massive amounts of heterogeneous data.
- Program taught entirely in English.

### M.S. in Statistics (*Diplôme d'ingénieur*)

ENSAI (NATIONAL SCHOOL FOR STATISTICS AND DATA ANALYSIS)

Rennes, France

Sept. 2014 - Oct. 2017

- Training topics: Statistics, Econometrics and Computer Science.

### Statistics of random processes

AARHUS UNIVERSITY

Aarhus, Denmark

Jan. 2016 - Jun. 2016

- ERASMUS exchange.

### CPGE MPSI/MP

LYCÉE CLEMENCEAU

Reims, France

Sep. 2011 - Jun. 2014

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## Skills

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- Programming** Python, **R**, Bash
- Tools** Google Cloud Platform, Git, Continuous Integration
- Languages** French (mother tongue), English (C1, fluent)

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## Teaching Experience

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### Data analysis methods

STT-7335

University Laval

Jan. 2026 - Apr. 2026

- Class size: around 40 students.

### Bayesian statistics

STT-7145

University Laval

Feb. 2026 - Mar. 2026

- Class size: around 40 students.

## Data analysis

STT-2200

- Link: [stt2200.netlify.app](https://stt2200.netlify.app)
- Class size: around 30 students.

University Laval

Sep. 2025 - Dec. 2025

## R for Statistical Data Science

MS6071

- Link: [r-programming-course.netlify.app](https://r-programming-course.netlify.app)
- Class size: around 25 students.

University of Limerick

Sep. 2023/24 - Dec. 2023/24

## Engineering maths 5

MA4006

- Vector fields and integration.

University of Limerick

Feb. 2023

## Duration models

M.S IN STATISTICS

- Theoretical and practical (with **R**) sessions.
- Class size: around 30 students.

Ensaï

Apr. 2021 - Jun 2021

## Linear regression

M.S IN STATISTICS

- Theoretical and practical (with **R**) sessions.
- Class size: around 30 students.

Ensaï

Sep. 2019 - Nov. 2021

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# Scientific production

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## ARTICLES

### Covariance estimation for derivatives of functional data using an additive penalty in P-splines

ZHU Y., GOLOVKINE S., SIMPKIN A. J. & BARGARY N.

- Link: [arXiv:2602.20029](https://arxiv.org/abs/2602.20029)

arXiv preprint

2026

### On the use of the Gram matrix for multivariate functional principal components analysis

GOLOVKINE S., GUNNING E., SIMPKIN A. J. & BARGARY N.

- DOI: [10.1016/j.jmva.2025.105525](https://doi.org/10.1016/j.jmva.2025.105525)

JMVA

2025

### Is Stephen Curry really a guard? New perspective on players typology using functional data analysis

GOLOVKINE S. AND GUNNING E.

- Link: [arXiv:2504.21761](https://arxiv.org/abs/2504.21761)

arXiv preprint

2025

### Adaptive optimal estimation of irregular mean and covariance functions

GOLOVKINE S., KLUTCHNIKOFF N. & PATILEA V.

- DOI: [10.3150/24-BEJ1759](https://doi.org/10.3150/24-BEJ1759)

Bernoulli

2025

### FDAPy: a Python package for functional data

GOLOVKINE S.

- DOI: [10.21105/joss.07526](https://doi.org/10.21105/joss.07526)

JOSS

2025

### On the estimation of the number of components in multivariate functional principal component analysis

GOLOVKINE S., GUNNING E., SIMPKIN A. J. & BARGARY N.

- DOI: [10.1080/03610918.2025.2459862](https://doi.org/10.1080/03610918.2025.2459862)

Commun. Stat. - Simul. Comput.

2025

### Analysing kinematic data from recreational runners using functional data analysis

GUNNING E., GOLOVKINE S., SIMPKIN A. J., BURKE A., DILLON S., GORE S., MORAN K., O'CONNOR S., WHYTE E. & BARGARY N.

- DOI: [10.1007/s00180-024-01591-1](https://doi.org/10.1007/s00180-024-01591-1)

Comput Stat

2025

## Derivative Estimation of Multivariate Functional Data

ZHU Y., GOLOVKINE S., SIMPKIN A. J. & BARGARY N.

- Link: [arXiv:2411.18398](https://arxiv.org/abs/2411.18398)

*arXiv preprint*

2024

## A multivariate multilevel longitudinal functional model for repeatedly observed human movement data

GUNNING E., GOLOVKINE S., SIMPKIN A. J., BURKE A., DILLON S., GORE S., MORAN K., O'CONNOR S., WHYTE E. & BARGARY N.

- Link: [arXiv:2408.08481](https://arxiv.org/abs/2408.08481)

*arXiv preprint*

2024

## Menstrual cycle and hormonal contraceptive phases' effect on elite rowers' training performance and wellness

ANTERO J., GOLOVKINE S., NIFFOI L., MEIGNIÉ A., CHASSARD T., DELAROCHELAMBERT Q., DUCLOS M., MAITRE C., MACIEJEWSKI H., DIRY A. & TOUSSAINT J.-F.

- DOI: [10.3389/fphys.2023.1110526](https://doi.org/10.3389/fphys.2023.1110526)

*Front. Physiol.*

2023

## Learning the smoothness of noisy curves with application to online curve estimation

GOLOVKINE S., KLUTCHNIKOFF N. & PATILEA V.

- DOI: [10.1214/22-EJS1997](https://doi.org/10.1214/22-EJS1997)

*EJS*

2022

## Clustering multivariate functional data using unsupervised binary trees

GOLOVKINE S., KLUTCHNIKOFF N. & PATILEA V.

- DOI: [10.1016/j.csda.2021.107376](https://doi.org/10.1016/j.csda.2021.107376)

*CSDA*

2021

## CONFERENCE PROCEEDINGS

### Functional multilevel modelling of the influence of the menstrual cycle on the performance of female cyclists

GOLOVKINE S., CHASSARD T., MEIGNIÉ A., BRUNET E., TOUSSAINT J.-F. & ANTERO J.

- Link: [Proceedings of the 37th International Workshop on Statistical Modelling](#)

*IWSM*

2023

### Clustering multivariate functional data using unsupervised binary trees

GOLOVKINE S., KLUTCHNIKOFF N. & PATILEA V.

- Link: [22nd European Young Statisticians Meeting - Proceedings](#)

*EYSM*

2021

### Lissage de données fonctionnelles par estimation de leur régularité locale

GOLOVKINE S., KLUTCHNIKOFF N. & PATILEA V.

- Link: [Proceedings of the 52èmes Journées de Statistiques de la Société Française de Statistique](#)

*JDS*

2020

## SOFTWARES

### pyspline (<https://github.com/StevenGolovkine/pyspline>)

PYTHON PACKAGE

### FDAPy (<https://github.com/StevenGolovkine/FDAPy>)

PYTHON PACKAGE

### denoisr (<https://github.com/StevenGolovkine/denoisr>)

R PACKAGE

### funestim (<https://github.com/StevenGolovkine/funestim>)

R PACKAGE

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## Conferences

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### WITH TALK

|           |                                                                                                |                         |
|-----------|------------------------------------------------------------------------------------------------|-------------------------|
| Dec. 2026 | <b>CFE-CMStatistics</b> , 20th International Joint Conference CFE-CMStatistics                 | Berlin, Germany         |
| Aug. 2026 | <b>ICES</b> , 9th International Conference on Econometrics and Statistics                      | Kyoto, Japan            |
| May 2026  | <b>SSC</b> , Statistical Society of Canada Annual Meeting                                      | Hamilton, Canada        |
| Apr. 2026 | <b>ASA</b> , Virtual Sports Analytics Conference: Rising Stars and All Stars                   | Virtual Conference      |
| Nov. 2025 | <b>Workshop</b> , Symposium en apprentissage statistique (invited)                             | Québec, Canada          |
| Nov. 2025 | <b>Séminaire</b> , Séminaire de statistique (invited)                                          | Québec, Canada          |
| Oct. 2025 | <b>STATQAM</b> , Séminaire STATQAM (invited)                                                   | Montréal, Canada        |
| May 2024  | <b>JDS</b> , 55rd Statistical Days                                                             | Bordeaux, France        |
| Sep. 2023 | <b>NSHPC</b> , National Sport and Human Performance Conference 2023                            | Limerick, Ireland       |
| Jul. 2023 | <b>IWSM</b> , 37th International Workshop on Statistical Modelling                             | Dortmund, Germany       |
| Jul. 2023 | <b>JDS</b> , 54rd Statistical Days                                                             | Brussels, Belgium       |
| May 2023  | <b>CASI</b> , 43rd Conference on Applied Statistics in Ireland                                 | Killarney, Ireland      |
| Aug. 2022 | <b>SDS</b> , CSDA & EcoSta Workshop on Statistical Data Science (invited)                      | Bologna, Italy          |
| Jun. 2022 | <b>JDS</b> , 53rd Statistical Days                                                             | Lyon, France            |
| Feb. 2022 | <b>YSP</b> , 10th Young Statisticians and Probabilists day (invited)                           | Virtual Conference      |
| Dec. 2021 | <b>CMStatistics</b> , 14th conference on Computational and Methodological Statistics (invited) | Virtual Conference      |
| Sep. 2021 | <b>EYSM</b> , 22nd European Young Statisticians Meetings (invited)                             | Virtual Conference      |
| Jun. 2021 | <b>JDS</b> , 52nd Statistical Days                                                             | Virtual Conference      |
| Mar. 2021 | <b>Mathematics Seminars</b> , Hunter College (invited)                                         | Virtual Conference      |
| Dec. 2020 | <b>CMStatistics</b> , 13th conference on Computational and Methodological Statistics (invited) | Virtual Conference      |
| Nov. 2020 | <b>StatMod2020</b> , Statistical Modeling with Applications (invited)                          | Virtual Conference      |
| Jun. 2019 | <b>JDS</b> , 51st Statistical Days                                                             | Nancy, France           |
| Mar. 2019 | <b>MASCOT-NUM</b> , Annual Conference                                                          | Rueil-Malmaison, France |

### ATTENDED

|           |                                                    |                |
|-----------|----------------------------------------------------|----------------|
| Jun. 2019 | <b>DS<sup>3</sup></b> , Data Science Summer School | Saclay, France |
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## Student supervision

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### POSTGRADUATE STUDENTS (PHD)

|      |                                                                                                |                        |
|------|------------------------------------------------------------------------------------------------|------------------------|
| 2023 | <b>Ansari Z. H., Moore J. &amp; Ndukwe R. N.</b> , Spatio-temporal models for disease mapping. | University of Limerick |
|------|------------------------------------------------------------------------------------------------|------------------------|

### GRADUATE STUDENTS (MSc, *diplôme d'ingénieur*)

|      |                                                                                                   |                        |
|------|---------------------------------------------------------------------------------------------------|------------------------|
| 2024 | <b>Abhilash Patade</b> , Bayesian analysis for football data.                                     | University of Limerick |
| 2023 | <b>McDermott H.</b> , The home advantage in rugby.                                                | University of Limerick |
| 2022 | <b>Harrington T.</b> , Bayesian analysis for sport data.                                          | University of Limerick |
| 2022 | <b>Scanlon S.</b> , Does grid position affect who wins the race in F1?                            | University of Limerick |
| 2020 | <b>Manaa A., Maissoro H. &amp; Samaila Z.</b> , Classification de courbes après recalage.         | ENSAI                  |
| 2020 | <b>Gervillie R., Li J.-U. &amp; Rousseaux F.</b> , Détection de points d'impact dans des courbes. | ENSAI                  |

### UNDERGRADUATE STUDENTS (BSc)

|      |                                                                                         |                  |
|------|-----------------------------------------------------------------------------------------|------------------|
| 2026 | <b>Fradette J.</b> , RKHS et apprentissage machine.                                     | Université Laval |
| 2026 | <b>Garnier-Gidas N.</b> , Prévion de l'évolution des dépenses militaires dans le monde. | Université Laval |